

Part 7 - Hardware Acceleration for Pyflate

1. Bottleneck and main idea

The slow part of Pyflate is not one large calculation. Instead, Python repeats a small bit-by-bit Huffman lookup for every decoded symbol. This happens about 148,000 times in one decompression, so we chose `HuffmanTable.find_next_symbol` as the target for hardware acceleration.

A local `cProfile` run of three decompressions took 4.101 seconds. Out of this, 1.667 seconds were spent in 444,813 calls to `find_next_symbol`, which is 40.6% of the runtime. A second run gave 42.9%. We use the lower value in the speedup calculation so that the estimate is not overly optimistic. The measurement still needs to be repeated in the course QEMU environment before submission.

Once the Python Huffman tables are converted into arrays, the work done by this function is fairly simple: keep the next bits from the compressed stream, find the first matching Huffman code range, return the symbol, and consume the bits that were used. These are small integer operations that can be implemented directly in hardware.

2. Accelerator design

We divided the accelerator into three SystemVerilog modules:

- 1 `bit_buffer.sv` receives compressed bytes and stores them in a 64-bit MSB-first buffer. It exposes the next 20 bits to the decoder and removes the number of bits used by each decoded symbol.
- 2 `huffman_decoder.sv` stores the Huffman tables and searches for a matching code. During the real-input test, we found that BZip2 can keep up to six Huffman groups and change the selected group every 50 symbols. The final design therefore keeps six table banks in hardware.
- 3 `pyflate_huffman_accelerator.sv` connects the two modules and exposes the external configuration and streaming signals.

For each possible code length L , the decoder performs the following check:

```
prefix = next_20_bits >> (20 - L)
match  = first_code[L] <= prefix <= last_code[L]
index  = first_symbol_index[L] + prefix - first_code[L]
```

If the range matches, `symbol_memory[index]` is returned and L bits are removed from the input buffer. If it does not match, the decoder checks the next length.

One option was to compare all possible lengths in parallel. That would improve throughput, but would also use more comparators and create a longer combinational path. We chose a simpler FSM that checks one length per clock cycle. It may need up to 20 checks, but the actual average for the supplied input is only 2.361 checks per symbol. It also avoids a very large 2^{20} -entry lookup table.

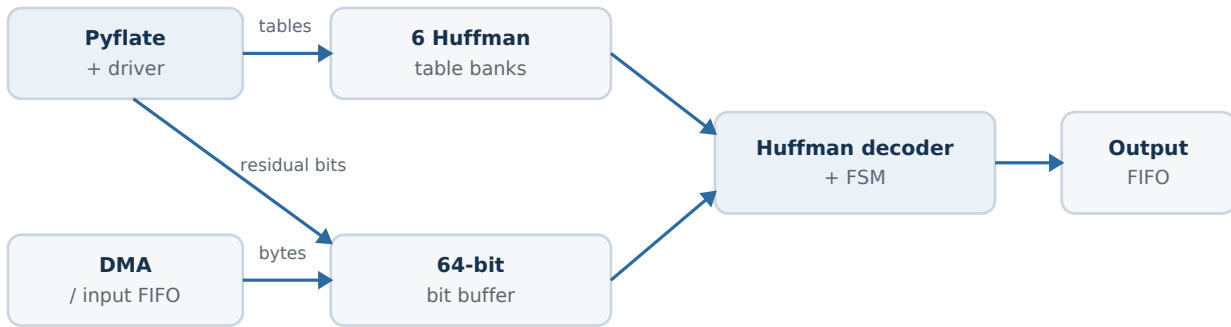


Figure 1. Proposed Pyflate hardware/software data flow

The RTL uses ready/valid handshakes. We use 100 MHz, or a 10 ns clock period, as the target frequency. This is a proposed target and was not confirmed by static timing analysis.

Interface	Signals	Width and purpose
Clock and reset	clk, rst_n, stream_reset	1 bit each; global reset and stream reset
Remaining bits	seed_valid, seed_ready, seed_data, seed_count	1-bit handshake, 64-bit data, 7-bit count
Byte input	byte_valid, byte_ready, byte_data	1-bit handshake and one 8-bit byte
Table selection	cfg_table_select, decode_table_select	3 bits; selects one of six banks
Table bounds	cfg_bounds_valid, cfg_min_length, cfg_max_length	1-bit valid and two 5-bit lengths
Length entry	cfg_length_valid, cfg_length, cfg_first_code, cfg_last_code, cfg_first_symbol_index	1, 5, 20, 20, and 9 bits
Symbol entry	cfg_symbol_valid, cfg_symbol_index, cfg_symbol_value	1, 9, and 9 bits
Decode request	decode_valid, decode_ready	1-bit ready/valid handshake
Decode result	symbol_valid, symbol_ready, symbol_value, bits_consumed	1-bit handshake, 9-bit symbol, 5-bit length
Status	cfg_ready, need_more_bits, decode_error, busy, buffered_bits	1-bit flags and a 7-bit buffer count

3. Connection to Pyflate

Pyflate still parses the BZip2 block header in software. It then converts the Huffman tables into the arrays expected by the accelerator and writes them into the six banks. The selector list that Pyflate already creates tells the hardware which bank to use for each group of symbols.

The handoff from software to hardware has one less obvious case. Header parsing can stop in the middle of a byte. At that point, RBitfield already contains a few unread bits while the file pointer has moved to the following byte. The driver must send these bits through seed_data and seed_count before sending the rest of the byte stream. In the supplied input, the handoff contained four bits: 1110.

The compressed bytes and decoded symbols should be moved in batches through FIFOs and DMA. Performing one MMIO access or interrupt per symbol would probably cost more than the lookup itself. From the software side, the driver needs operations similar to these:

```

configure_block(tables, selector_list, residual_bits, residual_count)
submit_compressed_buffer(input_address, byte_count)
receive_symbols(output_address, maximum_symbols) -> consumed_bit_count

```

DMA may read a few bytes beyond the final Huffman symbol, so the driver should continue from the logical position reported by `consumed_bit_count`, not from the physical read-ahead position. The move-to-front, run-length, BWT, CRC, and output-validation stages remain in software.

We implemented the accelerator interface itself, but not a specific AXI, DMA, or MMIO wrapper. Those parts depend on the SoC in which the accelerator is used.

4. Performance and hardware cost

We use Amdahl's law to estimate the effect on the complete benchmark. With $p = 0.406$ as the measured fraction and S_{hw} as the effective speedup of the Huffman lookup, including communication overhead:

$$S_{total} = 1 / ((1 - p) + p / S_{hw})$$

Huffman speedup	Estimated total speedup
2x	1.255x
4x	1.438x
8x	1.551x
Infinite	1.684x

The real input contains 148,271 Huffman symbols. Based on their code lengths, the FSM needs 646,642 modeled cycles, including request and output-handshake cycles. At 100 MHz this is about 6.47 ms before DMA and table-configuration overhead. If the hardware makes `find_next_symbol` effectively eight times faster, the complete benchmark should improve by about 1.55x. Even an ideal implementation cannot pass about 1.68x without moving more of the decompression pipeline to hardware.

The main area cost comes from keeping all six tables at the same time. Together they contain 20,292 bits of storage, in addition to the 64-bit input buffer. Yosys inferred seven memories with the expected 20,292 bits. Keeping only one table would save about 17 Kibit, but the table would have to be reloaded whenever the selector changes. We preferred to pay the memory cost and avoid that reconfiguration.

The iterative decoder also trades speed for area and power. It activates one comparison path per cycle, while a parallel decoder would activate several. Clock-gating the decoder while it is idle could further reduce switching. Exact area, frequency, and power numbers require a specific FPGA or ASIC target, so the current results should be treated as logical estimates rather than physical measurements.

5. Verification and remaining work

The directed testbench covers several code lengths, crossing a byte boundary, buffer underflow and refill, selection and reconfiguration of table banks, invalid input, and an invalid table number. It passes with both Icarus Verilog and Verilator and ends with `ALL TESTS PASSED`.

We also wanted to test more than a few manually selected codes. A Python script therefore records the Huffman tables, remaining input bits, compressed bytes, and expected results from an actual Pyflate run. The RTL then decodes the same trace. All 148,271 symbols and their code lengths matched the Python result, with zero mismatches. This test also checks all six table banks and the four-bit software-to-hardware handoff.

These tests verify the Huffman decoder and bit-buffer logic, but not a complete hardware/software system. The next step for a real SoC implementation would be to add the DMA/MMIO wrapper, pass the decoded symbols through the remaining software stages, and compare the final decompressed output with the existing MD5 value.